EV Chips: SiC/GaN Power Semiconductors, the Global Supply Chain and India's Position

Capstone 12: AI Research Report on the Semiconductor Industry · AI Skills Course (Topics 1-9) · Electronics and Communication Engineering
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1. Executive summary

An electric vehicle (EV) needs power chips that switch high voltages efficiently, battery-monitoring chips that keep the pack safe, and control chips that run the motor. This report asks how silicon carbide (SiC) and gallium nitride (GaN) compare with silicon, who controls the supply chain, and where India stands. The same three questions were put to four AI tools; every claim was then checked against government, company, research and news pages, and every number was recomputed by hand.

- **SiC beats silicon in the inverter, but the size of the gain depends on the test.** A simulation study reports over 40% lower conduction loss, over 60% lower switching loss and over 1% higher efficiency for a SiC MOSFET inverter against a Si-IGBT inverter [1]. ST claimed up to 3% in 2016 [8]. Each figure is quoted with its conditions.
- **Supply is concentrated.** Four companies held 82% of SiC substrate revenue in 2024 [11]. In 2023 ST led SiC power devices with 32.6%, and the top five suppliers together held 91.9% [38].
- **India is import-dependent but building.** NITI Aayog puts imports at 90-95% of semiconductor demand [17]. The Cabinet has approved 12 manufacturing projects worth over ₹1.64 lakh crore, including the first commercial SiC fab at Bhubaneswar [35][15][32], and approved ISM 2.0 on 15 July 2026 [35].
- **The risk picture is mixed.** Global SiC substrate revenue fell 9% in 2024 amid oversupply [11], yet automotive-grade parts may stay scarce [21].
- **Approved is not producing.** Fabs and packaging units still need yields, automotive qualification and customers.

2. Research design

2.1 Research questions

- **RQ1.** Which chips are used in an EV (power, battery management, motor control), and what do SiC and GaN offer over silicon in efficiency and cost?
- **RQ2.** Who controls the EV chip supply chain (Infineon, STMicroelectronics, onsemi and others), and what is India's position (ISM, Tata Electronics, CG Power)?
- **RQ3.** What are the main trends and risks (shortage, cost, import dependency)?

2.2 Method and AI workflow

1 Prompt 3 research questions → **2 Four AI tools** answers saved → **3 Split** into single claims → **4 Check** source + hand calculation → **5 Source log** status, counts, open actions → **6 Write** verified claims only

Each AI answer was split into single claims, and each claim got one verdict: Verified, Partly verified, Conflict, Unsupported or Not checked. Only facts from checked sources appear here; AI answers are never used as sources. Weak sources (exam-prep sites, broker blogs, vendor marketing blogs) were rejected. Every URL is in a spreadsheet source log with date, type, trust level and verification status; status counts and an open-action flag update automatically. After drafting, all cited sources were opened again on 24-25 October 2026 against the claim each one supports.

AI tool	What it produced	Citations	Main weakness found
Perplexity	Three structured tables	Site names only, no URLs	Of 20 claims checked: 9 verified, 3 partly, 1 conflict, 2 unsupported, 5 not checked; skipped India in RQ2 and the SiC advantage in RQ1

Claude	Answers with 27 numbered sources	Numbered, with URLs	No EDA/IP, foundry or ADAS coverage; many vendor sources; August 2025 status out of date
ChatGPT	Short Tanglish guide	5 official or company URLs	Few numbers; useful caution that approval is not output
Gemini	Technical report with comparison matrix	None	Many specific numbers with no source; one calculation disagrees with geometry

3. Findings

3.1 RQ1: EV chips and the SiC/GaN advantage

An EV uses several chip families with different jobs.

Table 1. Main chip groups in an EV

Subsystem	Main chips	Supporting evidence
Traction inverter and motor control	SiC MOSFETs or Si IGBTs, isolated gate drivers, real-time MCU, isolated bias supply	TI lists the MCU, isolated gate driver and isolated bias supply as key inverter components and says SiC FETs are widely adopted [2]
Battery management (BMS)	Cell-monitor analog front end, isolation and communication ICs, MCU	NXP's BMx7318 monitors 18 cells with isolated daisy-chain communication, up to ASIL C [3]; isolation sits between the high-voltage measurement side and the low-voltage MCU [26]; cell data feeds state-of-charge and state-of-health estimates [4]
On-board charger and DC-DC	PFC controller, DC-DC converter, GaN or SiC switches	A GaN-based 6.6 kW, 450 V bidirectional charger with an integrated 1 kW, 12 V converter has been reported [5]

Vehicle control, ADAS and cockpit chips (MCUs, SoCs, memory, network PHYs) were described by two AI tools but not independently verified, so they are outside the findings.

Table 2. SiC and GaN versus silicon: evidence found

Point	Evidence	Source and caution
Inverter losses	SiC MOSFET inverter: conduction loss down over 40%, switching loss down over 60%, efficiency up over 1% across the torque-speed curve	Simulation study [1], not a measurement
Vendor efficiency claim	Up to 3% for a 650 V SiC MOSFET against an equivalent IGBT	ST, 2016 [8]; different test from [1]
Size and cooling	A 220 kW SiC inverter had over 2,400 W lower losses at high torque (active switches only); cooling could shrink by over 30%; peak efficiency 99.1%	Single vendor example, ROHM, 2018 [7]
Temperature	Wide-bandgap devices are more efficient and tolerate higher temperatures than silicon	US DOE, qualitative [9]
Voltage fit	650 V GaN suits 400 V systems; 800 V systems favour 1,200 V SiC	Trade article [6]
Material properties	Bandgap: Si 1.12 eV, 4H-SiC 3.26 eV, GaN about 3.4 eV (2.9× Si for SiC). Thermal conductivity: Si 1.5, SiC 4.9 W/cm-K (3.3× Si); GaN 1.3 to 2.0 W/cm-K depending on source	Checked against two pages [39][40]; GaN value differs between them
Cost	SiC substrates are over 50% of a SiC device's bill of materials; higher efficiency can lower overall system cost	[23]; [22] (2021)

Reading the numbers. Reported efficiency gains range from about 1% (simulation [1]) to up to 3% (vendor claim [8]). Differences in test conditions explain the spread, so every quoted figure carries its source and year. A blog claim of 3-5% was rejected as unreliable.

Cost trade-off. SiC chips cost more, but a more efficient inverter can mean less cooling. Gemini claimed a 5-8% smaller battery and a 5-10% longer range; no checked source supports these figures, so they are not used. The sourced position is only that system-level savings can offset the higher device cost [22].

Where GaN fits. GaN suits compact chargers and DC-DC converters in the 650 V class [6]. The DOE draft projects GaN at about 3% of power electronics by 2028 and names cost as a limit [10], so GaN traction use is less mature than SiC. Figure 1 shows the DOE projection.

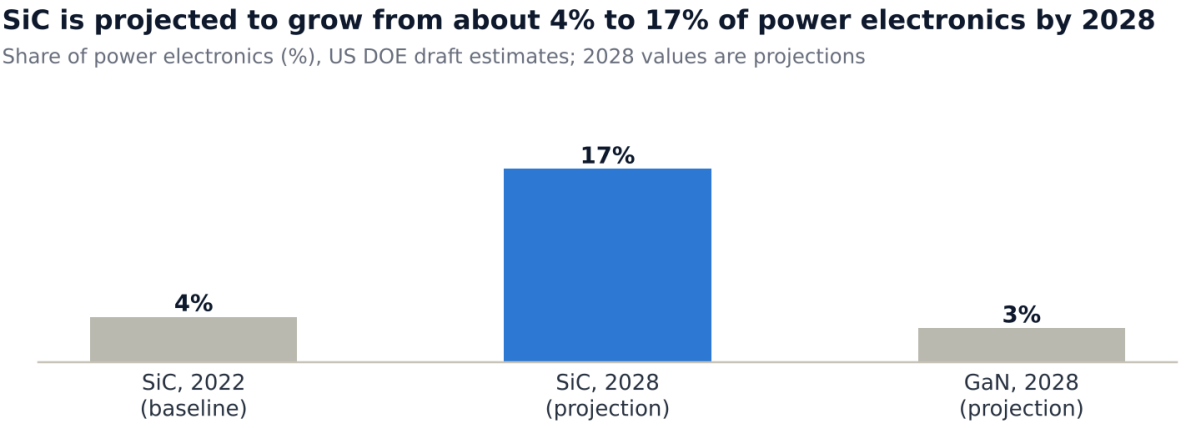


Figure 1. Source: US DOE, 2023 Critical Materials Assessment (draft) [10]. Projections, not measured data; no 2022 GaN baseline is given.

3.2 RQ2: Who controls the supply chain, and where is India?

Global position. The chain runs from chip design to wafer fabrication, packaging and testing, then the power module, inverter or BMS supplier and finally the automaker. SiC is concentrated at two levels. In substrates, Wolfspeed led in 2024 with 33.7%, ahead of TanKeBlue (17.3%) and SICC (17.1%), both Chinese, and Coherent (13.9%); the four hold 82.0% and the two Chinese vendors 34.4% [11]. The market was worth US$1.04 billion, down 9% on the year [11]. Figure 2 shows the shares.

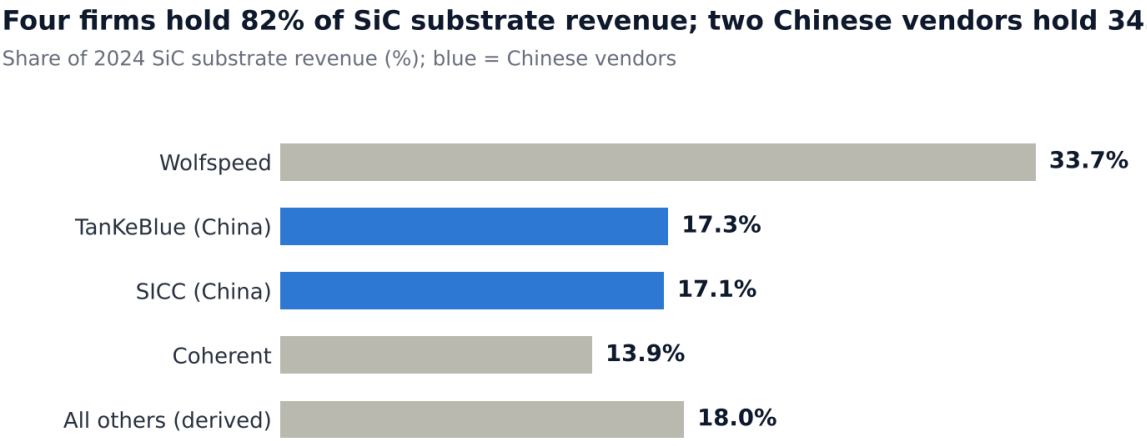


Figure 2. Source: TrendForce [11]. "All others" is derived as 100 minus 82.

In power devices, ST led in 2023 with 32.6% and onsemi rose to second; the top five suppliers held 91.9%, leaving 8.1% for all others (derived) [38]. onsemi meets over half of its own substrate need after buying GTAT, while Infineon has no in-house SiC crystal production [12]. Automotive discrete chips must pass the AEC-Q101 stress-test standard [13] and suppliers work under the IATF 16949 quality system [27]. Automotive use may need qualification beyond AEC-Q101 [25]. This is why new suppliers, including new Indian fabs, take time to win orders. No single qualification duration was found; an AI-supplied "12-18 months" had no source and was rejected.

India's position. The India Semiconductor Mission (ISM) was launched in 2021 with a ₹76,000 crore outlay [29]. By 15 July 2026 the Cabinet had approved 12 manufacturing projects with over ₹1.64 lakh crore of investment: one semiconductor fab, two compound-semiconductor fabs and nine packaging units [35][30]. Officials say two of the ten earlier projects have started commercial shipments and two more are expected soon [28]. ISM 2.0, announced in the 2026-27 Budget [30], was cleared by the Finance Ministry's expenditure department at about ₹1.25 lakh crore [31] and approved by the Cabinet on 15 July 2026 at about ₹1.27 lakh

crore [35]. On the design side, 24 projects are supported, 105 companies have been given design tools and 23 tapeouts are complete [30].

Table 3. Key Indian projects and their EV relevance

Project	Type	Verified detail	EV relevance
Tata Electronics + PSMC, Dholera	Wafer fab	₹91,000 crore; 50,000 wafer starts a month; 28 nm compute and power-management chips including for EVs [14]	Silicon logic fab, not a SiC power plant
Tata Semiconductor Assembly and Test, Morigaon (Assam)	Assembly and test	₹27,000 crore; 48 million chips a day; EV is a target segment [14]	Packaging capacity
CG Power + Renesas + Stars, Sanand	OSAT	₹7,600 crore and 15 million chips a day [14]; ₹7,584 crore in a Rajya Sabha reply [16] (same project, rounded)	Auto and industrial chips
Kaynes Semicon, Sanand	OSAT	₹3,307 crore [16]	Packaging
SiCSem + Clas-SiC Wafer Fab (UK), Bhubaneswar	SiC compound fab	Approved August 2025 as the first commercial SiC fab [15][36]; 60,000 wafers and 96 million packaged units a year [36]; operation expected 2027-28 [32]	**Most relevant to EV power chips**
Crystal Matrix (Dholera) and Suchi Semicon (Surat)	Two units approved May 2026	About ₹3,936 crore combined; Suchi capacity 1,033 million chips a year [28]	Not linked to EV traction in sources found

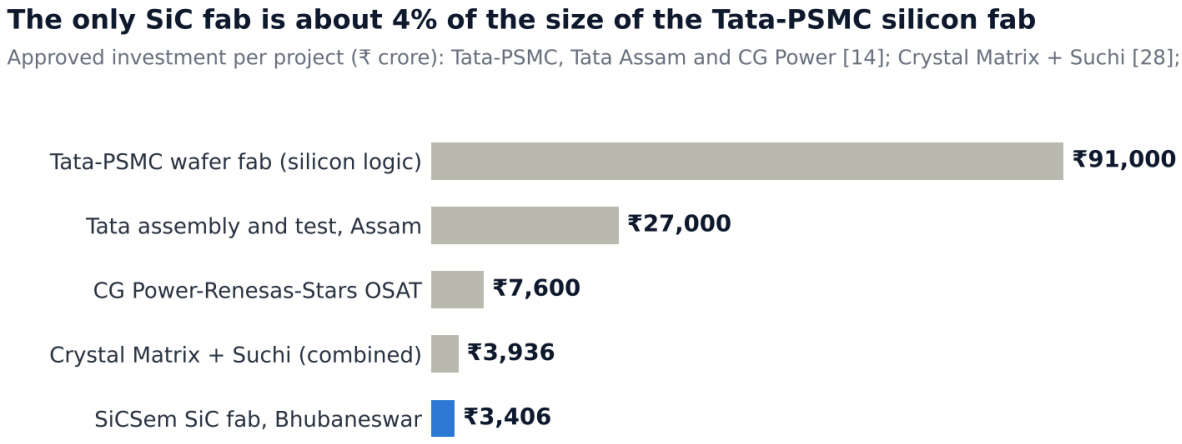


Figure 3. Sources: Tata-PSMC, Tata Assam, CG Power [14]; Crystal Matrix and Suchi combined [28]; SiCSem [33]. A silicon fab dwarfs the one SiC project (3,406 / 91,000 = 3.7%).

SiCSem output check. A September 2026 approval by the Falta SEZ Unit Approval Committee lists 4.8 million SiC diodes and 91.2 million SiC MOSFETs a year, which sum to 96 million and match the 96 million packaged units announced earlier [34][36]. SiC MOSFETs are the main EV inverter device. The investment figure is **unresolved**: about ₹2,000 crore (US$220 million) was reported at ground-breaking in early November 2025 [32], while the September 2026 approval cites ₹3,406.25 crore [33][34]. The scopes may differ, so both are quoted with sources.

National goals. NITI Aayog's May 2026 roadmap targets 35-50% of domestic chip demand and a US$120-150 billion value chain by 2035 [18], and proposes two compound-semiconductor fabs for SiC and GaN [17].

3.3 RQ3: Trends and risks

Table 4. Main trends and risks

Trend or risk	Evidence	Impact
Import dependency	India imports about 90-95% of semiconductor demand [17]; NITI's US$240 billion import figure for 2035 is a scenario if the trend continues, not a forecast [17]	Foreign-exchange outflow and supply disruption

SiC adoption growing	DOE draft: EVs are the main driver of SiC demand; SiC rises from about 4% of power electronics in 2022 to about 17% by 2028 (projection) [10]	Rising demand for devices and substrates
Oversupply and price pressure	Substrate revenue fell 9% in 2024 with oversupply expected [11]; Renesas temporarily suspended in-house SiC development amid weak EV demand and cheaper Chinese supply [20]	Price volatility; weaker suppliers exit
Automotive-grade parts may stay tight	McKinsey expects usable automotive-grade wafer supply to be significantly lower than announced capacity because of strict defect limits (forecast) [21]; DOE calls SiC wafer production slow, energy-intensive and low-yield [10]	Headline oversupply can hide a shortage of qualified parts
Cost	SiC substrates exceed 50% of device bill of materials [23]	Slows SiC in price-sensitive EVs
Geopolitics and materials	China required export licences for gallium and germanium in July 2023, added antimony in August 2024 and banned exports of the three to the US in December 2024 [37]; Chinese vendors hold 34.4% of substrates [11]	GaN material risk and supply concentration
Qualification time	AEC-Q101 plus extra qualification for automotive use [13][25]	New suppliers earn orders slowly
India execution	First plants rely heavily on imported equipment, materials and technology; about US$80 billion of further incentives may be needed to 2035 (industry estimate) [24]	Self-reliance will be gradual

Hand checks. 33.7 + 17.3 + 17.1 + 13.9 = 82.0%; 17.3 + 17.1 = 34.4%; 100 − 91.9 = 8.1%; ISM 2.0 against ISM 1.0 is 127,500 ÷ 76,000 − 1 = 67.8%; SiCSem 4.8 + 91.2 = 96 million units. All match the sources or are labelled derived.

4. Analysis: what the evidence means

Trends. SiC is already the main wide-bandgap choice for traction inverters, GaN is growing in chargers and DC-DC, and EV demand is the main driver of SiC growth [2][6][10]. Higher-voltage 800 V platforms favour 1,200 V SiC [6].

Risks. Three matter most. First, concentration: a few companies control substrates and devices, and Chinese vendors are gaining share [11][38]. Second, mismatch: general oversupply can coexist with scarcity of automotive-grade parts [11][21]. Third, execution: India's approvals must become yields, qualified products and customers [14][24].

India's realistic opportunity. The near-term openings are packaging and testing, mature-node and power chips, chip design and local supplier development. The SiCSem fab is the project most closely linked to EV power devices, with operation expected in 2027-28 [32]. Because approved projects do not guarantee volumes or dates, targets such as 35-50% self-sufficiency [18] should be read as goals.

5. Where the AI tools disagreed

Topic	What the tools and sources said	Resolution
SiC substrate: shortage or oversupply	Perplexity listed both a bottleneck and oversupply; Gemini stressed shortage	TrendForce 2024 shows oversupply [11]; qualified automotive parts may still be tight [21]
Efficiency gain over IGBT	Over 1% [1], up to 3% [8], 3-5% (blog)	Quoted [1] and [8] with conditions; rejected the blog
Number of ISM projects	10 (Aug 2025), 12, "six" (blog)	Resolved: 12 approved units, over ₹1.64 lakh crore, per the 15 July 2026 release [35]
Qualification time	"12-18 months" (Perplexity)	No source; rejected [25]
200 mm SiC wafers	Gemini: 1.4× more dies	Area ratio (200 ÷ 150)² = 1.78×, so 1.4× is not supported; not used
onsemi and Infineon device shares	23.6% and 16.5% appeared in AI output	Not found on the TrendForce pages [38][12]; removed

SiCSem investment	About ₹2,000 crore versus ₹3,406 crore	Unresolved; both quoted with sources [32][33]
GaN thermal conductivity	1.3 W/cm-K (Gemini); 2.0 on another page	Sources differ: 1.3 [40], 2.0 [39]; range given

The full prompt library, hallucination log and verification report are in the Submission Pack.

6. Responsible use, copyright and limitations

Copyright and citation. All source content is paraphrased and each fact points to a numbered source; direct quotation was avoided. AI output was treated as a lead to check, never as evidence. Only public information was used and no private or confidential material was uploaded. Derived numbers are labelled derived; projections and scenarios are labelled as such.

Limitations. (1) Three sources, [5], [16] and [23], could not be re-opened in the final check, so their claims are supporting detail only. (2) Market-share data covers 2023-2024 and may have changed. (3) Some sources are company or trade articles; their numbers are treated as claims. (4) The SiCSem investment figure is unresolved. (5) EDA/IP concentration, advanced logic foundries, ADAS chips and India's design-talent claims were not researched in depth. (6) The efficiency study [1] is a simulation, not a measurement.

7. Conclusion

SiC is the leading material for EV traction inverters, delivering lower losses than silicon, with a cost penalty that system savings may offset. Supply is concentrated in a few firms and Chinese vendors are growing quickly. India's 12 approved projects, including a planned SiC fab, are real progress, but import dependency of 90-95% will fall only as plants reach production and win automotive qualification. The practical lesson from the AI comparison is that every tool made useful drafts and every tool needed checking: invented numbers, outdated claims, mismatched citations and missing sources all appeared.

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